

App Dev Project Report - TREKTHEHILLS

1. Student Details

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About Me:

I am a student in the IIT Madras BS Degree Program with a strong interest in web application development and database systems. I enjoy building practical, user-friendly applications that solve real-world problems.

2. Project Details

Project Title: TrekTheHills - Trekking Management System

Problem Statement:

Trekking organizers typically coordinate treks, staff assignments, and participant bookings through scattered spreadsheets and manual communication, making it hard to track slot availability, staff approvals, and booking history reliably. TrekTheHills is a web-based application that centralizes trek publishing, staff assignment, and booking management under one system with clear role boundaries.

Approach:

The app was built using Flask with a blueprint-based modular structure, splitting functionality across `auth`, `admin`, `staff`, and `user` blueprints. Admins create and manage treks, approve or blacklist staff and users, and oversee all bookings. Staff manage only the treks assigned to them, which is updating slots, advancing trek status, and viewing participants. Users browse open treks, book available slots, and manage their own bookings and profile. Role-based access is enforced with custom decorators, and session state is managed through Flask-Login.

3. AI/LLM Declaration

The extent of AI/LLM usage was approximately 10%, limited to debugging assistance and documentation. All core application logic like routes, booking rules, and role-based access control was implemented and integrated manually.

4. Technologies and Frameworks Used

Technology / Library	Purpose
Flask 3.0	Core backend web framework
Flask-SQLAlchemy	Object Relational Mapper for the SQLite database

Technology / Library	Purpose
Jinja2	Template engine for rendering dynamic HTML pages
Bootstrap 5 + Bootstrap Icons	Frontend styling, components, and responsive layout
Flask-Login	Session-based authentication and access control
Werkzeug Security	Password hashing and verification
SQLite	Lightweight local relational database

5. Database Schema / ER Diagram

Tables:

User - stores account details for all roles (`id`, `username`, `email`, `full_name`, `password_hash`, `role`, `phone`, `is_blacklisted`, `is_approved`, `created_at`)

Trek - stores trek listing details (`id`, `name`, `location`, `difficulty`, `duration_days`, `altitude`, `total_slots`, `available_slots`, `status`, `start_date`, `end_date`, `description`, `price`, `assigned_staff_id`, `created_at`)

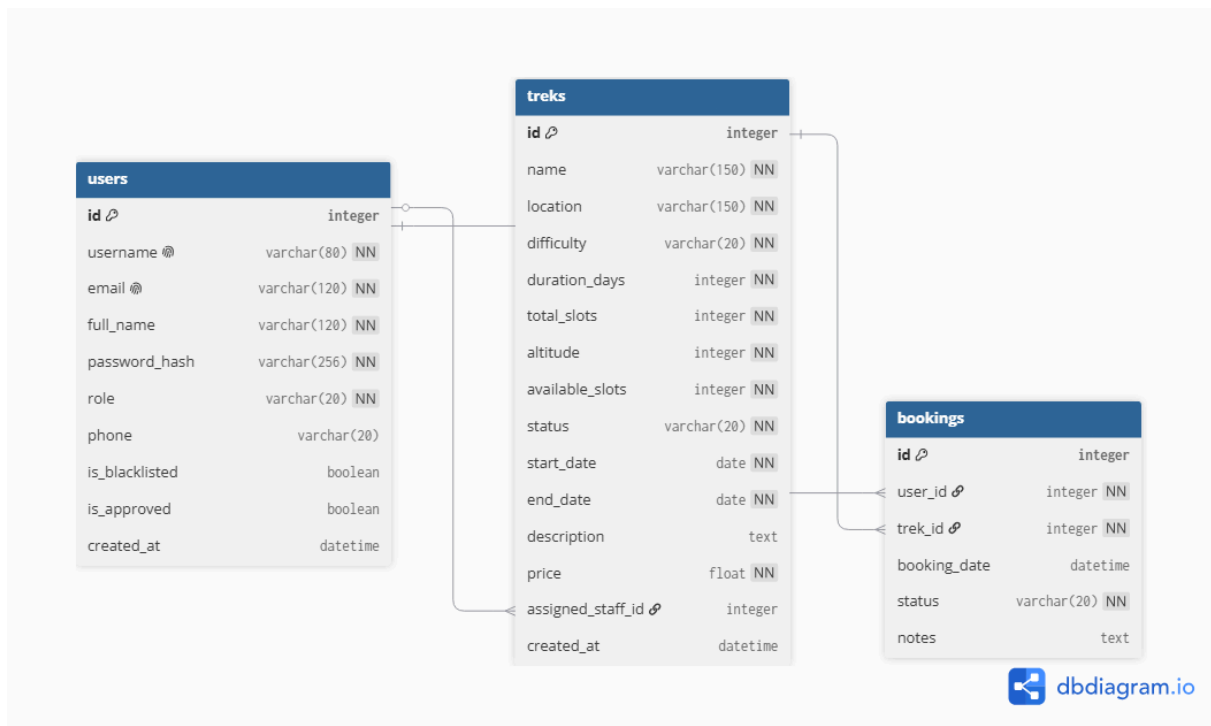
Booking - stores booking records (`id`, `user_id`, `trek_id`, `booking_date`, `status`, `notes`)

Relationships:

One-to-Many → User (staff) → Trek

One-to-Many → User → Booking

One-to-Many → Trek → Booking



6. Application Routes

Route	Method	Description
/login	GET, POST	Authenticate a user and redirect by role
/register	GET, POST	Register a new trekker (auto-approved)
/staff/register	GET, POST	Register as staff (requires admin approval)
/admin/treks/add	GET, POST	Create a new trek and optionally assign staff
/admin/treks/<id>/edit	GET, POST	Edit trek details, slots, status, and staff assignment
/staff/trek/<id>	GET, POST	Manage an assigned trek's slots and status
/user/treks	GET	Browse open treks with search/filter
/user/trek/<id>/book	POST	Book an available slot on a trek
/user/booking/<id>/cancel	POST	Cancel an active booking

7. Architecture and Features

Architecture Overview: `app.py` — application factory, blueprint registration, default admin seeding

`models.py` — SQLAlchemy models (`User` , `Trek` , `Booking`)

`auth.py` / `admin.py` / `staff.py` / `user.py` — Flask blueprints per role

`decorators.py` — role-based access-control decorators

`templates/` — Jinja2 templates organized by role (`admin/` , `staff/` , `User/` , `auth/`)

`static/css/` — modular stylesheets (navbar, sidebar, cards, forms, dashboard, utilities)

Implemented Features:

- Role-based login for Admin, Staff, and User with blacklist/approval checks
- Trek creation, editing, and deletion with staff assignment
- Staff-managed slot updates and trek status progression
- Booking system with duplicate-booking prevention and cancellation
- Participant list view per trek for assigned staff
- Dashboards summarizing key stats per role
- Responsive UI built with Bootstrap 5

Additional Features:

- Profile management (name, email, phone, password) for both users and staff
- Few attributes (Max elevation for different treks and experience level for trekkers)

8. Video Presentation

Video Drive Link: <https://drive.google.com/file/d/1mZ9N55FKb-wWtujYEdFyHgFyMRaZbzaY/view?usp=sharing>